What's next?

After the despair, grief, and confusion, I think it's time for us to try and be constructive to figure out what math will be like in the new era.

Let me try to work through the process and the questions that would naturally arise, using the recent OAI drop as an example.

Trying to be an optimist, the natural thing to do would be to read the OAI files and try to understand them, and then write nice expository papers about them and maybe post them on arxiv or another repository, or even submit them to a journal.

We most likely all agree that such an expository effort would be a noble undertaking and would count for something, but the big question is: how much exactly do we want to reward a mathematician for that?

For instance, would such an expository paper be worth as much as a paper with new results in the pre-AI era? Would such a paper be publishable?

A common answer might be that this could be publishable provided that the author has contributed something new; however, in general, how can we measure what is "new" now that some researchers get new results by having LLMs produce them?

This is something that only the math community can answer, but I think it is important to start proposing concrete answers to such questions.

Of course, to do such an expository job nicely would take some time, let's say 1-2 months. This obviously raises some other issues:

1. another mathematician (or even an amateur) can ask the models to further improve on the OAI paper and produce a "better" result, for instance improving the bounds. This way, you would be preparing an exposition of a result that is already "old".

2. another mathematician can write another expository account of the same result. Whose paper then would we value as the one "of record"? Would it matter who is first one to post the paper, or who writes the best exposition?

3. Also, let's say a grad student has been working on one such conjecture, and their conjecture is proven by AI. Clearly they will be devastated, but at least you could tell them: "well, maybe you can write a nice explanation of their result and how it relates to your partial progress". Yet, if both a young mathematician and a well-established master write such an exposition, wouldn't we be more naturally inclined to take the established person's account in higher regard? This fact alone, or even the risk this might happen, might likely dissuade young researchers to try doing that.

Then also come the ethical and political issues: in particular, I am fairly sure that several of our colleagues think that any exegesis of the slopdrop would amount to "collaborating with the enemy", as in doing the job that OAI should be doing instead; according to this more radical view, mathematicians should "resist" and wait until the tech companies write up their results nicely and submit them to journals. This is also a valid point of view, even though in my opinion I don't think AI labs will ever do that.

I certainly do not have prescriptions about all these norms, but I think it is time to seriously discuss them in our community as mathematicians. Here, I would like to point out the important role of senior mathematicians: given their power in the community, it is on them to take responsibility and go ahead trying to figure this out.

It is our duty to address these issues, and try to reach some consensus, lest we completely lose the grip of current mathematical research to techbros and button-pushers.

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